

Key Concepts: Properties & Concentrations of Solutions

- **Solubility Factors:** The ability of a solute to dissolve depends on the nature of the solvent and solute. Key factors influencing the *rate* of dissolving include stirring, temperature, and surface area. Solubility is generally expressed as grams of solute per 100 g of solvent.
- **Concentrations:** To find Molarity (M), divide moles of solute by liters of solution. Remember, dilution does not change the total moles of solute, only the concentration.
- **Colligative Properties:** These properties (vapor-pressure lowering, freezing-point depression, and boiling-point elevation) depend on the number of solute particles present, not their identity. The magnitude of these effects is directly proportional to the number of solute particles.
- **Advanced Concentrations:** Molality (m) and mole fractions are additional ways to express concentration. Note that freezing-point depression and boiling-point elevation are directly proportional to molality.

Vocabulary Definitions

- **Solution Terms:**
 - **Miscible/Immiscible:** Describes liquids that dissolve in each other (miscible) or do not (immiscible).
 - **Concentrated/Dilute Solution:** A solution with a large amount of solute (concentrated) versus a small amount (dilute).
 - **Saturated/Unsaturated/Supersaturated:** The state of a solution regarding its ability to dissolve more solute at a given temperature.
- **Quantitative Terms:**
 - **Concentration:** A measure of the amount of solute in a given amount of solvent or solution.
 - **Molarity (M):** Moles of solute per liter of solution.
 - **Molality (m):** Moles of solute per kilogram of solvent.
 - **Mole fraction (X):** The ratio of moles of one component to the total moles in the mixture.
 - **Solubility:** The maximum amount of substance that will dissolve at a specific temperature.
- **Colligative Properties:**
 - **Colligative Property:** A property that depends only on the number of solute particles.
 - **Boiling-point elevation:** The increase in boiling point caused by adding solute.
 - **Freezing-point depression:** The decrease in freezing point caused by adding solute.

Formulas:

Formula	Equation
Henry's Law	$\frac{S_1}{P_1} = \frac{S_2}{P_2}$
Molarity	$M = \frac{\text{moles of solute}}{\text{liters of solution}}$
Dilution	$M_1 \times V_1 = M_2 \times V_2$
Percent by Volume	$\frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$
Percent by Mass	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$
Molality	$m = \frac{\text{moles of solute}}{\text{kilogram of solvent}}$
Mole Fractions	$X_A = \frac{n_A}{n_A + n_B}$
Freezing-point Depression	$\Delta T_f = i \times K_f \times m$
Boiling-point Elevation	$\Delta T_b = i \times K_b \times m$

4. Practice Problems

1. **Henry's Law:** The solubility of a gas is 0.5 g/L at 2 atm. What is the solubility at 4 atm (assuming temperature is constant)?
2. **Molarity:** What is the molarity of a solution containing 2.0 moles of NaCl in 0.5 liters of solution?
3. **Dilution:** You have 100 mL of 6.0 M HCl. What volume of water must be added to create a 2.0 M solution?
4. **Percent by Mass:** What is the percent by mass of 25 g of sugar dissolved in 225 g of water?
5. **Molality:** What is the molality of a solution containing 0.5 moles of solute in 250 g (0.25 kg) of solvent?
6. **Boiling-point Elevation:** If a solution has a molality (m) of 2.0 and the K_b for water is $0.51\text{ }^{\circ}\text{C/m}$, what is the boiling point elevation (ΔT_b)?

5. Answer Key

1. 1.0 g/L
2. 4.0 M
3. 200 mL total volume
4. 10%
5. 2.0 molal
6. 1.02 °C